

11.4 Gram Matrix



Core skills you will master in this section:

- ▶ The Gram matrix is a symmetric matrix – positive semidefinite, or positive definite.
- ▶ The relationship between the Gram matrix and the inner products of the original matrix's column vectors.
- ▶ The main diagonal elements are the squared L2 norms of the column vectors.
- ▶ Spectral decomposition of the Gram matrix.
- ▶ Using an orthogonal matrix to orthogonalize the original matrix.
- ▶ Judging the "explanatory power" of eigenvectors from the size of the eigenvalues, and understanding their relative importance.
- ▶ The relationship between the spectral decomposition of the Gram matrix and singular value decomposition.

In the world of linear algebra and data science, there is a role that seems simple yet is absolutely central – the Gram matrix (Gram matrix). As mentioned earlier in this book, the Gram matrix is a special type of symmetric matrix.

Simple as its definition looks, the Gram matrix's influence is everywhere. From inner products, vector norms, spectral decomposition, to singular value decomposition; from positive definiteness to Cholesky decomposition and LDL decomposition; to the data matrices, covariance matrices, and correlation-coefficient matrices commonly seen in data analysis; and even to kernel methods in machine learning algorithms (such as Gaussian-kernel support vector machines and kernel principal component analysis) – all of these are, in some form, closely tied to the Gram matrix.

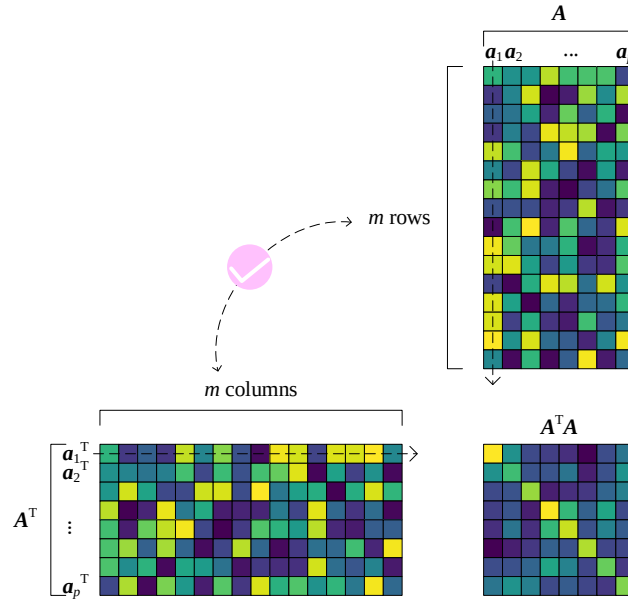
Given the importance of the Gram matrix, we devote an entire section to discussing it.

Definition

Given that the shape of matrix A is $m \times p$ – that is, A has m rows and p columns – A 's Gram matrix is

$$G = A^T A$$

As shown in Figure 1, A 's Gram matrix has shape $p \times p$.

Figure 1. Matrix $\mathbf{A}$'s Gram Matrix $\mathbf{A}^T \mathbf{A}$

Clearly, the Gram matrix $\mathbf{G}$ is a symmetric matrix

$$\mathbf{G}^T = (\mathbf{A}^T \mathbf{A})^T = \mathbf{A}^T \mathbf{A} = \mathbf{G}$$

In addition, the Gram matrix is a positive semidefinite matrix (positive semidefinite), that is, for any nonzero p -dimensional column vector $\mathbf{x}$, we always have

$$\mathbf{x}^T \mathbf{G} \mathbf{x} = \mathbf{x}^T \mathbf{A}^T \mathbf{A} \mathbf{x} = (\mathbf{A} \mathbf{x})^T (\mathbf{A} \mathbf{x}) = \|\mathbf{A} \mathbf{x}\|^2 \geq 0$$



Chapter 13, Section 1 of this book will explain positive definiteness in detail.

The determinant of the Gram matrix $\mathbf{A}^T \mathbf{A}$ is zero if and only if the column vectors of matrix $\mathbf{A}$ are linearly dependent.

$\mathbf{A}^T \mathbf{A}$'s rank equals $\mathbf{A}$'s column rank. This can be understood as: $\mathbf{A}^T \mathbf{A}$ merely "compresses and organizes" the inner-product relationships among the original column vectors, without increasing or decreasing the information's dimensionality.

Expanding $\mathbf{A}^T \mathbf{A}$ using the first perspective on matrix multiplication, you will see inner products.

The First Perspective on Matrix Multiplication

Using the first perspective on matrix multiplication, expand $\mathbf{A}^T \mathbf{A}$

$$\mathbf{A}^T \mathbf{A} = \begin{bmatrix} \mathbf{a}_1^T \\ \mathbf{a}_2^T \\ \vdots \\ \mathbf{a}_p^T \end{bmatrix} @ \begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_p \end{bmatrix} = \begin{bmatrix} \mathbf{a}_1^T \mathbf{a}_1 & \mathbf{a}_1^T \mathbf{a}_2 & \cdots & \mathbf{a}_1^T \mathbf{a}_p \\ \mathbf{a}_2^T \mathbf{a}_1 & \mathbf{a}_2^T \mathbf{a}_2 & \cdots & \mathbf{a}_2^T \mathbf{a}_p \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{a}_p^T \mathbf{a}_1 & \mathbf{a}_p^T \mathbf{a}_2 & \cdots & \mathbf{a}_p^T \mathbf{a}_p \end{bmatrix}$$

The equation above can also be written in vector inner-product form

$$\mathbf{A}^T \mathbf{A} = \begin{bmatrix} \mathbf{a}_1 \cdot \mathbf{a}_1 & \mathbf{a}_1 \cdot \mathbf{a}_2 & \cdots & \mathbf{a}_1 \cdot \mathbf{a}_p \\ \mathbf{a}_2 \cdot \mathbf{a}_1 & \mathbf{a}_2 \cdot \mathbf{a}_2 & \cdots & \mathbf{a}_2 \cdot \mathbf{a}_p \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{a}_p \cdot \mathbf{a}_1 & \mathbf{a}_p \cdot \mathbf{a}_2 & \cdots & \mathbf{a}_p \cdot \mathbf{a}_p \end{bmatrix} = \begin{bmatrix} \langle \mathbf{a}_1, \mathbf{a}_1 \rangle & \langle \mathbf{a}_1, \mathbf{a}_2 \rangle & \cdots & \langle \mathbf{a}_1, \mathbf{a}_p \rangle \\ \langle \mathbf{a}_2, \mathbf{a}_1 \rangle & \langle \mathbf{a}_2, \mathbf{a}_2 \rangle & \cdots & \langle \mathbf{a}_2, \mathbf{a}_p \rangle \\ \vdots & \vdots & \ddots & \vdots \\ \langle \mathbf{a}_p, \mathbf{a}_1 \rangle & \langle \mathbf{a}_p, \mathbf{a}_2 \rangle & \cdots & \langle \mathbf{a}_p, \mathbf{a}_p \rangle \end{bmatrix}$$

As shown in Figure 2, the main diagonal elements of the Gram matrix $\mathbf{A}^T \mathbf{A}$ are $\mathbf{A}$'s column vector inner product with itself, which can also be written as the squared L^2 norm, that is

$$\mathbf{a}_j^T @ \mathbf{a}_j = \mathbf{a}_j \cdot \mathbf{a}_j = \|\mathbf{a}_j\|^2$$

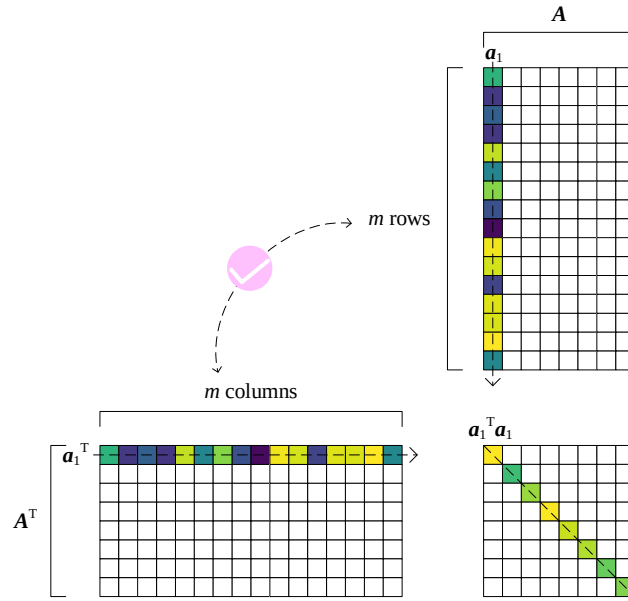


Figure 2. Matrix $\mathbf{A}$'s Gram Matrix – Diagonal Elements

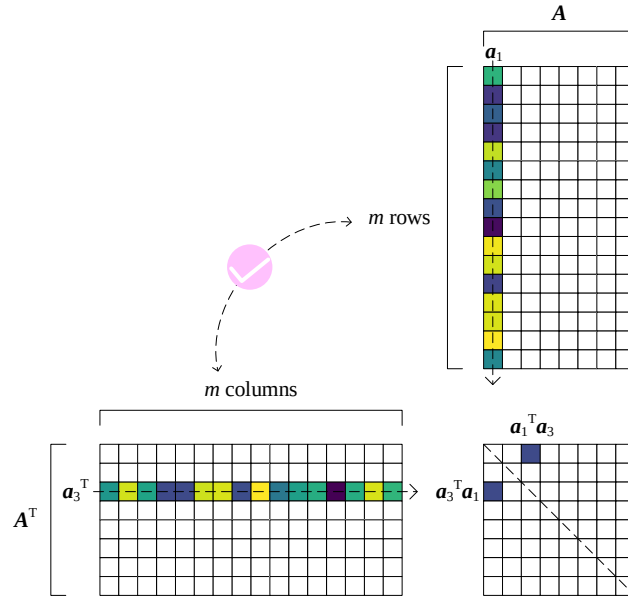
As shown in Figure 3, $\mathbf{A}^T \mathbf{A}$'s off-diagonal elements are $\mathbf{A}$'s pairwise inner products of the column vectors. For example,

$$\mathbf{a}_3^T \mathbf{a}_1 = \mathbf{a}_1^T \mathbf{a}_3$$

Looking at the equation above from the perspective of inner products,

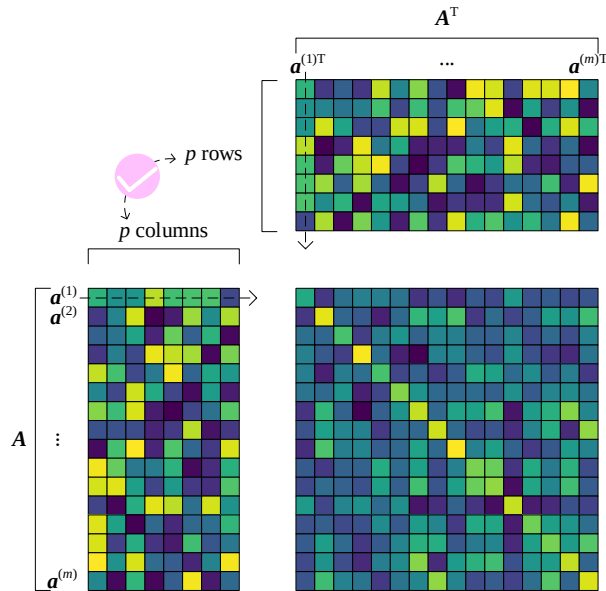
$$\mathbf{a}_1 \cdot \mathbf{a}_3 = \mathbf{a}_3 \cdot \mathbf{a}_1$$

This also confirms that $\mathbf{A}^T \mathbf{A}$ is a symmetric matrix.

Figure 3. Matrix A 's Gram Matrix – Off-Diagonal Elements

The Second Gram Matrix

It is worth noting that, as shown in Figure 4, AA^T is also a Gram matrix. AA^T 's main diagonal is A 's row-vector inner product with itself (squared L^2 norm); AA^T 's off-diagonal elements are A 's pairwise inner products of the row vectors. Clearly, AA^T is also a symmetric matrix.

Figure 4. Matrix A^T 's Gram Matrix AA^T

Spectral Decomposition of the Gram Matrix

As shown in Figure 5, performing spectral decomposition on $G = A^T A$ gives

$$G = V \Lambda V^T$$

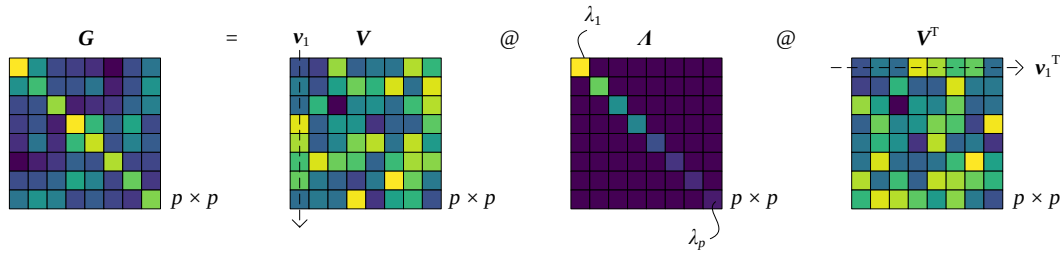
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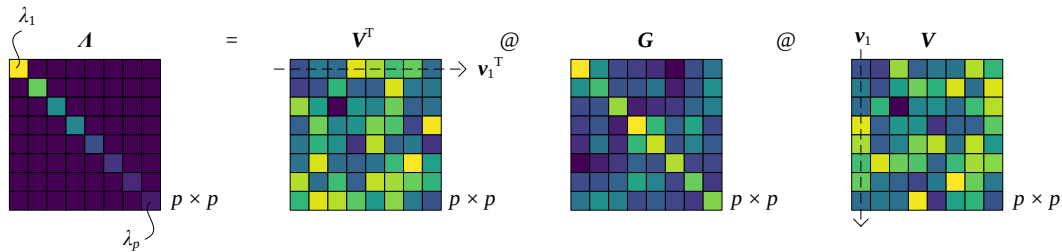
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Figure 5. The Gram Matrix $\mathbf{A}^T \mathbf{A}$ – Spectral Decomposition

Diagonalization

As shown in Figure 6, $\mathbf{G}$ is diagonalized to give

$$\mathbf{V}^T \mathbf{G} \mathbf{V} = \mathbf{A}$$

Figure 6. The Gram Matrix $\mathbf{A}^T \mathbf{A}$ – Diagonalization

Writing $\mathbf{V}$ as $[\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p]$, the equation above expands to

$$\underbrace{\begin{bmatrix} \mathbf{v}_1^T \mathbf{G} \mathbf{v}_1 & \mathbf{v}_1^T \mathbf{G} \mathbf{v}_2 & \cdots & \mathbf{v}_1^T \mathbf{G} \mathbf{v}_p \\ \mathbf{v}_2^T \mathbf{G} \mathbf{v}_1 & \mathbf{v}_2^T \mathbf{G} \mathbf{v}_2 & \cdots & \mathbf{v}_2^t \mathbf{G} \mathbf{v}_p \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{v}_p^T \mathbf{G} \mathbf{v}_1 & \mathbf{v}_p^T \mathbf{G} \mathbf{v}_2 & \cdots & \mathbf{v}_p^T \mathbf{G} \mathbf{v}_p \end{bmatrix}}_{\mathbf{V}^T \mathbf{G} \mathbf{V}} = \underbrace{\begin{bmatrix} \lambda_1 & & & \\ & \lambda_2 & & \\ & & \ddots & \\ & & & \lambda_p \end{bmatrix}}_{\mathbf{A}}$$

Figure 7 shows an example of computing the main diagonal elements of the diagonal matrix $\mathbf{A}$.

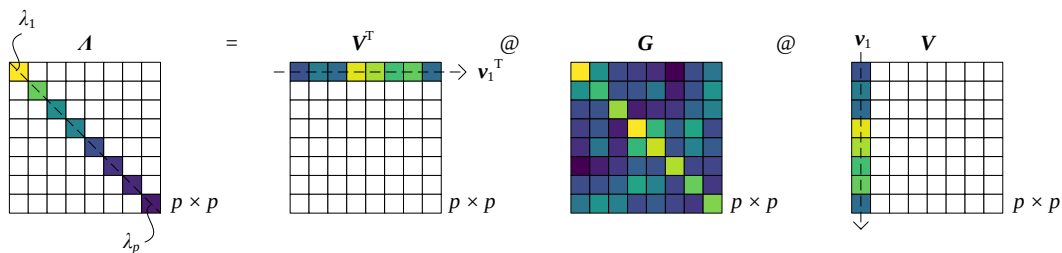
Figure 7. The Diagonal Matrix $\mathbf{A}$ – Main Diagonal Elements

Figure 8 shows an example of computing the off-diagonal elements of the diagonal matrix $\mathbf{A}$.

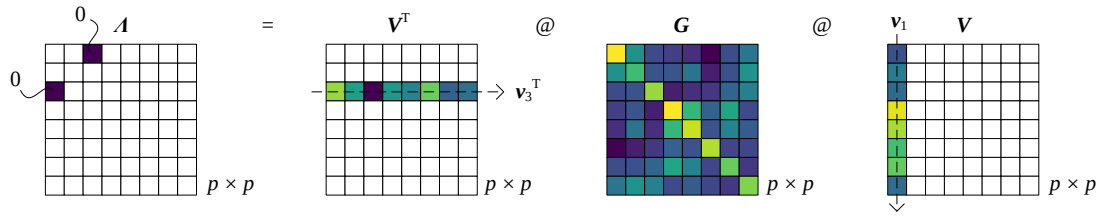
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Figure 8. The Diagonal Matrix $\mathbf{A}$ — Off-Diagonal Elements

Substituting $\mathbf{G} = \mathbf{A}^T \mathbf{A}$ into the equation above and expanding, we get

$$\underbrace{\begin{bmatrix} \mathbf{v}_1^T \mathbf{A}^T \mathbf{A} \mathbf{v}_1 & \mathbf{v}_1^T \mathbf{A}^T \mathbf{A} \mathbf{v}_2 & \cdots & \mathbf{v}_1^T \mathbf{A}^T \mathbf{A} \mathbf{v}_p \\ \mathbf{v}_2^T \mathbf{A}^T \mathbf{A} \mathbf{v}_1 & \mathbf{v}_2^T \mathbf{A}^T \mathbf{A} \mathbf{v}_2 & \cdots & \mathbf{v}_2^T \mathbf{A}^T \mathbf{A} \mathbf{v}_p \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{v}_p^T \mathbf{A}^T \mathbf{A} \mathbf{v}_1 & \mathbf{v}_p^T \mathbf{A}^T \mathbf{A} \mathbf{v}_2 & \cdots & \mathbf{v}_p^T \mathbf{A}^T \mathbf{A} \mathbf{v}_p \end{bmatrix}}_{\mathbf{v}^T \mathbf{G} \mathbf{v}} = \underbrace{\begin{bmatrix} \lambda_1 & & & \\ & \lambda_2 & & \\ & & \ddots & \\ & & & \lambda_D \end{bmatrix}}_{\mathbf{A}}$$

Simplifying the equation above, we get

$$\underbrace{\begin{bmatrix} (\mathbf{A} \mathbf{v}_1)^T (\mathbf{A} \mathbf{v}_1) & (\mathbf{A} \mathbf{v}_1)^T (\mathbf{A} \mathbf{v}_2) & \cdots & (\mathbf{A} \mathbf{v}_1)^T (\mathbf{A} \mathbf{v}_p) \\ (\mathbf{A} \mathbf{v}_2)^T (\mathbf{A} \mathbf{v}_1) & (\mathbf{A} \mathbf{v}_2)^T (\mathbf{A} \mathbf{v}_2) & \cdots & (\mathbf{A} \mathbf{v}_2)^T (\mathbf{A} \mathbf{v}_p) \\ \vdots & \vdots & \ddots & \vdots \\ (\mathbf{A} \mathbf{v}_p)^T (\mathbf{A} \mathbf{v}_1) & (\mathbf{A} \mathbf{v}_p)^T (\mathbf{A} \mathbf{v}_2) & \cdots & (\mathbf{A} \mathbf{v}_p)^T (\mathbf{A} \mathbf{v}_p) \end{bmatrix}}_{\mathbf{v}^T \mathbf{G} \mathbf{v}} = \underbrace{\begin{bmatrix} \lambda_1 & & & \\ & \lambda_2 & & \\ & & \ddots & \\ & & & \lambda_D \end{bmatrix}}_{\mathbf{A}}$$

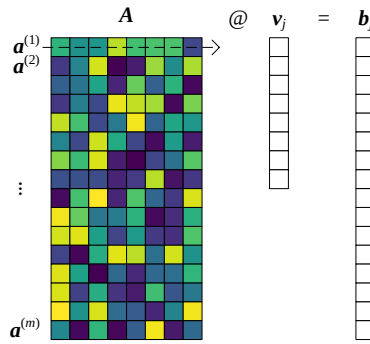
Projection

Let $\mathbf{b}_j = \mathbf{A} \mathbf{v}_j$ ($j = 1, \dots, p$), the equation above can be written as

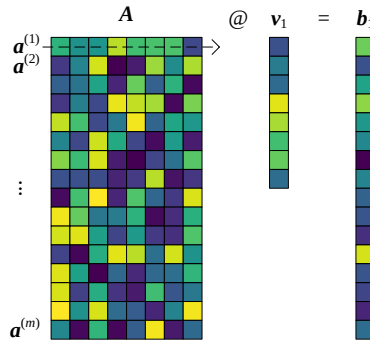
$$\underbrace{\begin{bmatrix} \mathbf{b}_1^T \mathbf{b}_1 & \mathbf{b}_1^T \mathbf{b}_2 & \cdots & \mathbf{b}_1^T \mathbf{b}_p \\ \mathbf{b}_2^T \mathbf{b}_1 & \mathbf{b}_2^T \mathbf{b}_2 & \cdots & \mathbf{b}_2^T \mathbf{b}_p \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{b}_p^T \mathbf{b}_1 & \mathbf{b}_p^T \mathbf{b}_2 & \cdots & \mathbf{b}_p^T \mathbf{b}_p \end{bmatrix}}_{\mathbf{B}^T \mathbf{B}} = \underbrace{\begin{bmatrix} \lambda_1 & & & \\ & \lambda_2 & & \\ & & \ddots & \\ & & & \lambda_p \end{bmatrix}}_{\mathbf{A}}$$

As shown in Figure 9, since $\mathbf{v}_j$ is a unit column vector, the matrix product $\mathbf{A} \mathbf{v}_j$ is equivalent to the projection of matrix $\mathbf{A}$ onto $\text{span}(\mathbf{v}_j)$, giving the result $\mathbf{b}_j$.

Each row vector of matrix $\mathbf{A}$ is equivalent to a data point.

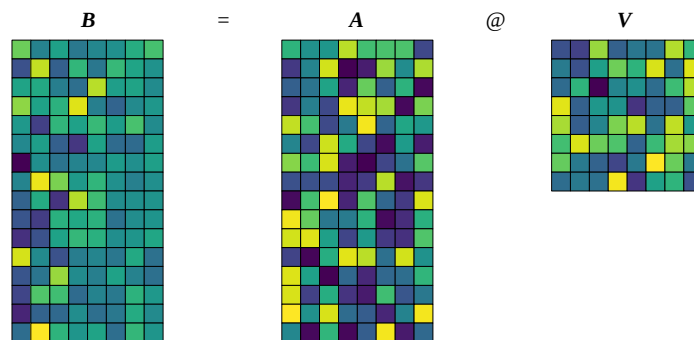
Figure 9. The Projection Result of Matrix A onto $\text{span}(v_j)$ Is b_j

For example, Figure 10 shows the projection of matrix A onto $\text{span}(v_1)$, giving the result b_1 .

Figure 10. The Projection Result of Matrix A onto $\text{span}(v_1)$ Is b_1

Orthogonalization

Figure 11 shows the matrix multiplication $B = AV$. Geometrically, the projection of matrix A onto $\text{span}(v_1, v_1, \dots, v_p)$ gives the result B .

Figure 11. The Projection Result of Matrix A onto $\text{span}(v_1, v_1, \dots, v_p)$ Is B

Substituting $B = AV$ into the equation above, we get

$$\mathbf{B}^T \mathbf{B} = \underbrace{\begin{bmatrix} \lambda_1 & & \\ & \lambda_2 & \\ & & \ddots \\ & & & \lambda_p \end{bmatrix}}_{\mathbf{A}}$$

As shown in Figure 12, the equation above is equivalent to the Gram matrix of $\mathbf{B}$.

Have you noticed that $\mathbf{B}$'s column vectors are pairwise orthogonal?!

In other words, $\mathbf{B} = \mathbf{A}\mathbf{V}$ is equivalent to completing the orthogonalization of matrix $\mathbf{A}$.

➔ The QR decomposition introduced earlier in this book is also an orthogonalization — please review it.

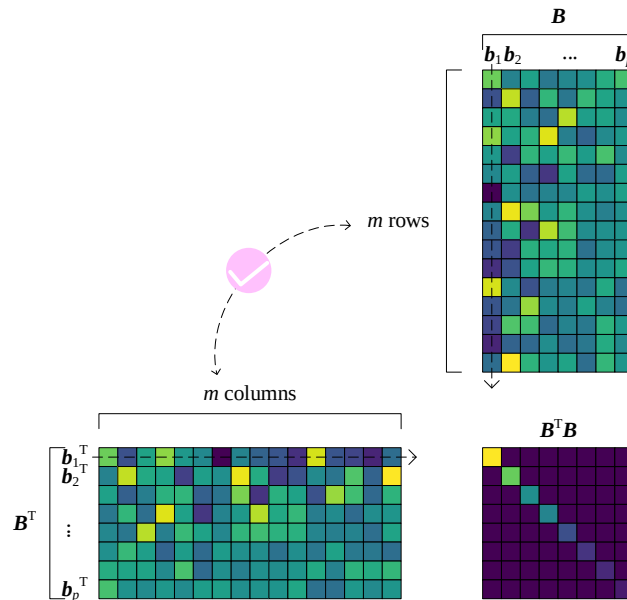
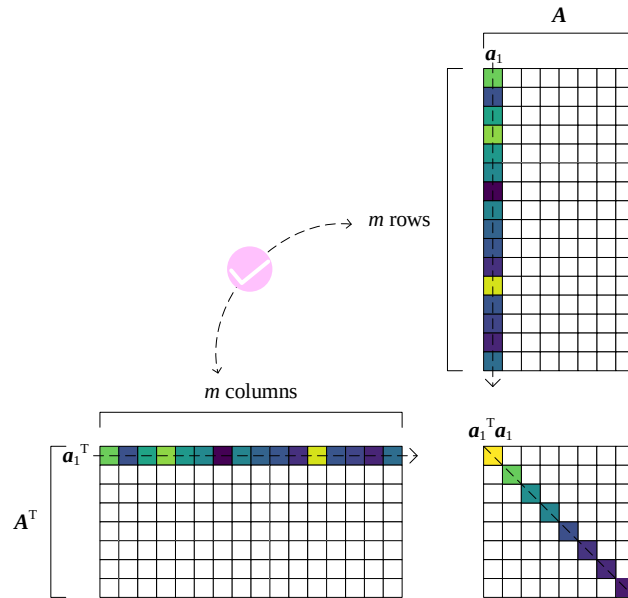


Figure 12. Matrix $\mathbf{B}$'s Gram Matrix $\mathbf{B}^T \mathbf{B}$

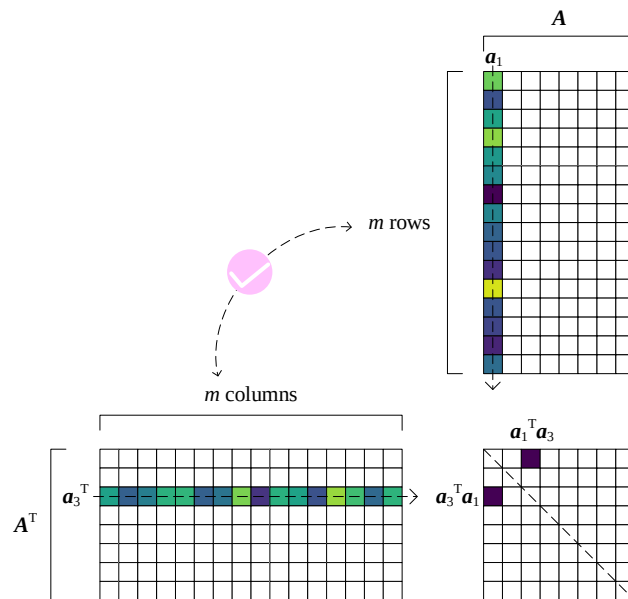
As shown in Figure 13, the diagonal elements of the equation above can clearly be written as the squared L^2 norm:

$$\|\mathbf{b}_j\|_2^2 = \|\mathbf{A}\mathbf{v}_j\|_2^2 = \lambda_j$$

⚠ Note that whenever you see a matrix product resulting in a scalar, you should always consider whether that matrix product can be written as an L^2 norm.

Figure 13. Matrix A 's Gram Matrix — Diagonal Elements

As shown in Figure 14, matrix A 's Gram matrix has off-diagonal elements equal to 0.

Figure 14. Matrix A 's Gram Matrix — Off-Diagonal Elements

Geometric Perspective

How should we understand the equation above?

Once again, let's bring out our specialty — the geometric perspective.

As shown in Figure 15, A 's row vectors are each a scatter point $\bullet$. These scatter points $\bullet$, projected onto $\text{span}(\mathbf{v})$, give the result $\mathbf{b}_j$ — that is, the $\bullet$ on the line in the figure. Each value in $\mathbf{b}_j$ is the distance from the corresponding $\bullet$ on the line to the origin.

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Figure 15 – the yellow • background points represent matrix A 's i -th row vector $a^{(i)}$. $a^{(i)}$'s projection onto v_j gives $b_j^{(i)}$, which is exactly $a^{(i)}$'s coordinate in $\text{span}(v_j)$:

$$b_j^{(i)} = a^{(i)} v_j$$

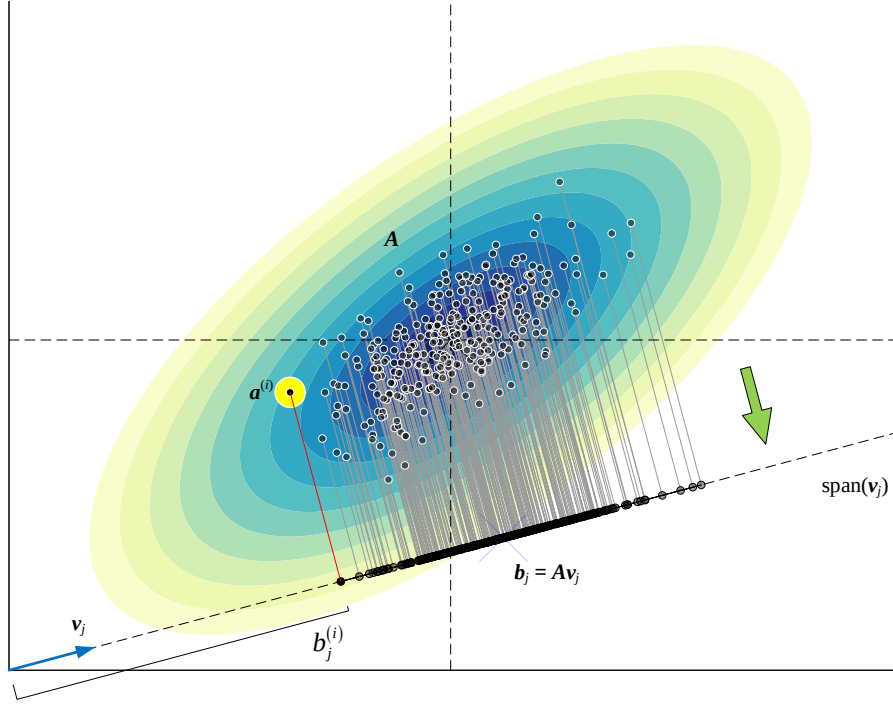


Figure 15. A Orthogonally Projected onto $\text{span}(v_j)$ Gives b_j – Geometric Perspective



Figure 15 – the ellipse in this figure is actually another kind of distance metric – the Mahalanobis distance. This is a topic covered in Chapter 13, Section 4.

With this perspective, we now know that in the equation above, $\|b_j\|_2^2$ represents the sum of squared distances (which can be positive or negative) from $b_j^{(i)}$ to the origin, that is:

$$\|b_j\|_2^2 = (b_j^{(1)})^2 + (b_j^{(2)})^2 + \dots + (b_j^{(n)})^2 = \lambda_j$$

The sum of squares of these distances happens to equal the eigenvalue λ_j .

If, in the equation above, the eigenvalues λ_j are arranged in decreasing order, that is $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_D$, this shows that the eigenvectors v_j also have a primary-secondary distinction.

As shown in Figure 16, matrix A is projected onto different eigenvectors v_j and the resulting $\|b_j\|_2^2 = \|Av_j\|_2^2$ values vary in size, with the size corresponding to the eigenvalue. This also means the eigenvectors also have a primary-secondary importance distinction. Among the infinitely many $\mathbb{R}^D$ in v_j , A projected onto the first eigenvector v_1 gives the largest $\|b_1\|_2^2 = \|Av_1\|_2^2$, with the maximum value being λ_1 .



This is exactly the core idea behind Principal Component Analysis, which will be introduced in Chapter 12 of this book.

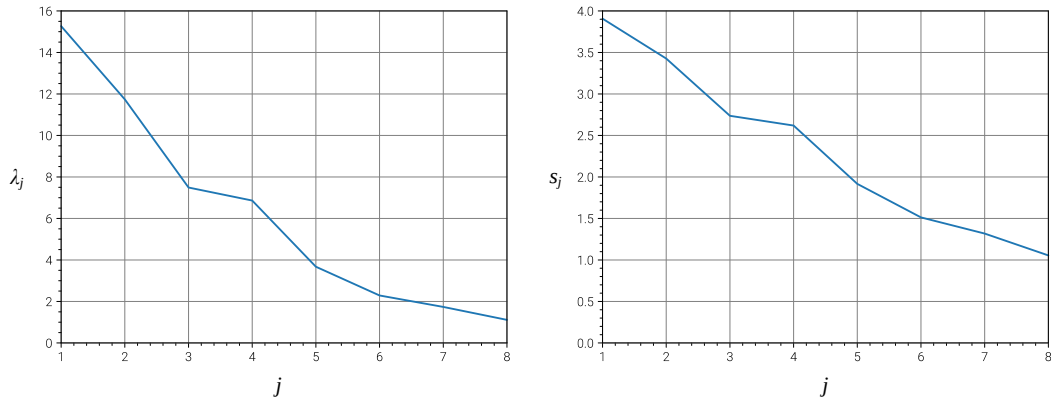


Figure 16. Eigenvalues and s_j (Square Roots of the Eigenvalues) Determine the "Explanatory Power" of the Eigenvectors

Normalization

Taking the square root of the equation above, we get

$$\|\mathbf{b}_j\| = \sqrt{(b_j^{(1)})^2 + (b_j^{(2)})^2 + \dots + (b_j^{(n)})^2} = \sqrt{\lambda_j}$$

Let $\|\mathbf{b}_j\| = \sqrt{\lambda_j} = s_j$, the equation above can be written as

$$\|\mathbf{b}_j\| = s_j$$

If the eigenvalue is greater than 0, we can normalize $\mathbf{b}_j$

$$\mathbf{u}_j = \frac{\mathbf{b}_j}{s_j}$$

In other words

$$\mathbf{b}_j = s_j \mathbf{u}_j$$

Substituting the equation above into $\mathbf{b}_j = \mathbf{A}\mathbf{v}_j$ we get

$$\mathbf{A}\mathbf{v}_j = s_j \mathbf{u}_j$$

Layer-by-Layer Superposition

Since $\mathbf{V}$ is an orthogonal matrix, matrix $\mathbf{A}$ can be written as

$$\begin{aligned}
 \underset{I}{A} &= AVV^T = A \begin{bmatrix} \mathbf{v}_1 & \mathbf{v}_2 & \cdots & \mathbf{v}_p \end{bmatrix} \begin{bmatrix} \mathbf{v}_1^T \\ \mathbf{v}_2^T \\ \vdots \\ \mathbf{v}_p^T \end{bmatrix} \\
 &= A \underbrace{(\mathbf{v}_1\mathbf{v}_1^T + \mathbf{v}_2\mathbf{v}_2^T + \cdots + \mathbf{v}_p\mathbf{v}_p^T)}_I \\
 &= A\mathbf{v}_1\mathbf{v}_1^T + A\mathbf{v}_2\mathbf{v}_2^T + \cdots + A\mathbf{v}_p\mathbf{v}_p^T
 \end{aligned}$$

In the equation above, $\mathbf{v}_j\mathbf{v}_j^T$ are all projection matrices!

Substituting the equation above into the equation above, and simplifying

$$\begin{aligned}
 A &= s_1\mathbf{u}_1\mathbf{v}_1^T + s_2\mathbf{u}_2\mathbf{v}_2^T + \cdots + s_p\mathbf{u}_p\mathbf{v}_p^T \\
 &= \sqrt{\lambda_1}\mathbf{u}_1\mathbf{v}_1^T + \sqrt{\lambda_2}\mathbf{u}_2\mathbf{v}_2^T + \cdots + \sqrt{\lambda_p}\mathbf{u}_p\mathbf{v}_p^T
 \end{aligned}$$

In other words, we have written matrix A as a layer-by-layer superposition of several matrices of the same shape, and s_j (the square root of the eigenvalue) determines the "explanatory power" of the eigenvectors, as shown specifically in Figure 16.

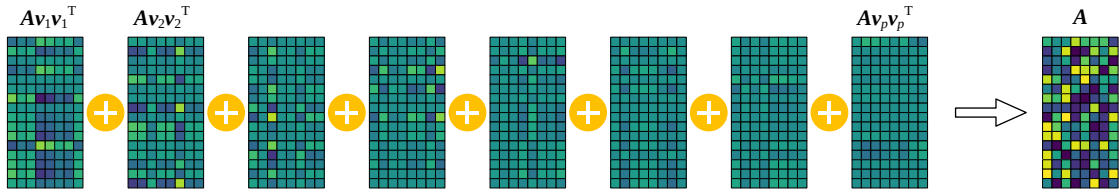


Figure 17. Layer-by-Layer Superposition

A Brand-New Matrix Decomposition

When j takes integer values from 1 to p , the equation above corresponds to a series of matrix multiplications

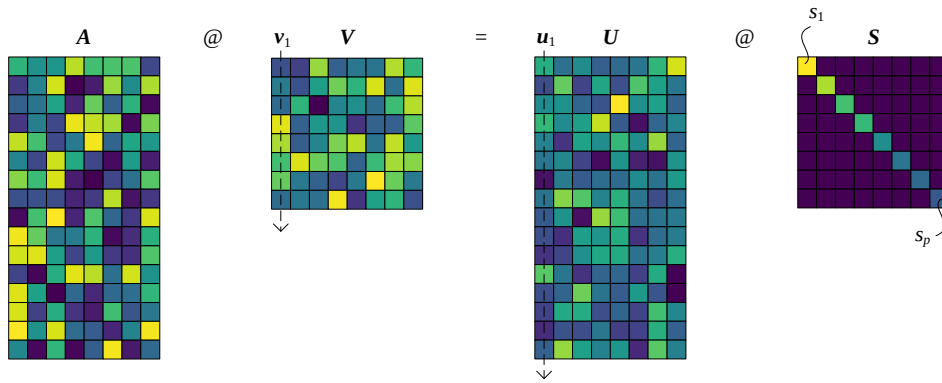
$$\begin{aligned}
 A\mathbf{v}_1 &= s_1\mathbf{u}_1 \\
 A\mathbf{v}_2 &= s_2\mathbf{u}_2 \\
 &\vdots \\
 A\mathbf{v}_p &= s_p\mathbf{u}_p
 \end{aligned}$$

Writing this series of multiplications as

$$A \begin{bmatrix} \mathbf{v}_1 & \mathbf{v}_2 & \cdots & \mathbf{v}_p \end{bmatrix} = \begin{bmatrix} \mathbf{u}_1 & \mathbf{u}_2 & \cdots & \mathbf{u}_p \end{bmatrix} \begin{bmatrix} s_1 & & & \\ & s_2 & & \\ & & \ddots & \\ & & & s_p \end{bmatrix}$$

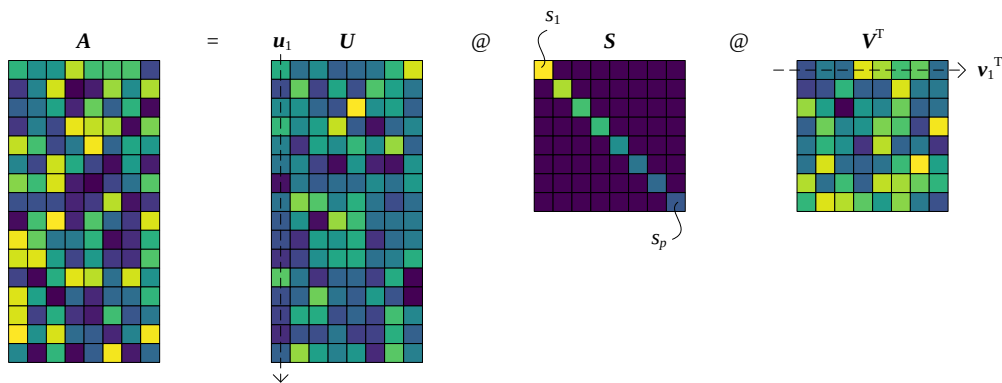
that is

$$AV = US$$

Figure 18. $AV = US$

Since V is an orthogonal matrix, V 's inverse equals its transpose, so the equation above can be written as

$$A = USV^T$$

Figure 19. Matrix A 's SVD

This actually leads us to the singular value decomposition covered in Chapter 14 of this book, as shown in Figure 19. S 's main diagonal elements are exactly the singular values. You will soon find that matrix U 's column vectors are also all unit vectors, and pairwise orthogonal!

In practical applications, there is an even more common type of Gram matrix — the covariance matrix — which is what we will introduce in the next chapter.



Please use tools such as DeepSeek/ChatGPT to complete the following exercises for this section.

Q1. Please compute the Gram matrix of the following matrix


$$\begin{bmatrix} 0 & 1 \\ 1 & 1 \\ 1 & 0 \end{bmatrix}$$


$$\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$$

Q2. Perform spectral decomposition on the Gram matrix from Q1.